MPJ Assignment :- 01

Name :- Sayali Patil
PRN :- 1032233901
Roll no :- 26
Panel :- D (D1)

PROBLEM STATEMENT :- Write a Java program to showcase classes, object creation, methods, constructions using student info.

THEORY :-

1] Define what is class & object with example.

⇒ Class :- It is a blueprint or template used to create objects. It defines the properties (variables) & behaviors (methods) that an object will have.
Eg.- In a student class, it will contain details like name, roll no, & a method to display them

Object :- It is a real world instance of a class. It represents an actual entity created using a class.
Eg.- If we have a student class and we create a student named 'Sayali', it will be an object.

2] Give the syntax to define methods in a class with examples for parameter passing & returning a value from methods.

⇒ Syntax :-

```
returnType methodName (parameter)
{
    // body
}
```

Eg.- With parameter passing

```
int add (int a, int b)
{
   return a+b;
}
```

Eg.- of returning value

```
int square (int n)
{
   return n*n;
}
```

3] Write about the use of constructor with example.

⇒ A constructor is a special method of a class that is automatically called when an object is created. It's main purpose is to initialize the data members of a class.

Use :-

i] Initialize object data

ii] Reduce extra code

iii] Ensures the object is ready to use immediately after creation.

eg.- class student

```
{
   string name;
   int rollno;
   Student (string n, int r) {
    name = n;
    rollno = r;
   }
}
```

FAQs :-

1] What is a class & object?

⇒ Class :- It is a blueprint that defines the properties & behaviours of an object. It tells what data an object will have & what action can it perform

object :- It is an instance of a class. It represents a real world entity created using a class.

Eg.- If student is a class, then student name & roll no is an object of that class.

2] What is a constructor in Java?

⇒ It is a special method that is automatically called when an object of a class is created. It is mainly used to initialize the data members of the class. The constructor has the same name as the class & doesn't have a return type. It ensures that the object is initialized at the time of creation.

3] What is the default access specifier and what is it's scope?

⇒ It is used when no access specifier is mentioned explicitly. It's scope is within the same package only. This means that the class members can be accessed by other classes in the same package but not from outside the package.

4] What are the different access specifiers?

→ Java provides four types of access specifiers :-

i] Private :- accessible only within the same class.

ii] Default :- accessible within the same package.

iii] Protected :- accessible within the same package & subclasses.

iv] Public :- accessible from anywhere.